

From Text to Large-n Datasets

Documenting the underlying subjectivities that shape dataset generation

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Observational Datasets

Large-*n* observational datasets are essential tools for social science.

> Used when experiments are **impractical, unhelpful, or unethical**, including for political phenomena like conflict, rights violations, democracy.

Dataset construction is an inevitably subjective process.

However, this inevitable subjectivity is often **unaddressed or unacknowledged.**

→ Resulting datasets produce a **misleading appearance of certainty**, and the prevailing treatment among downstream users is as if datasets are **objective**, **replicable**, and **definitive.**

Research questions

1. What strategies can be adopted to recognize, incorporate, and mitigate this subjective process?

2. Can a language model reproduce—or stand in for—the collective human coding process?

Framework

Sources of subjectivity for dataset construction

- **Underlying sources** are incomplete, contain contesting accounts, and over-represent mainstream viewpoints or actors able/willing to record events.
- **Research design** reflects the research director's assumptions and priorities
  - → Scope conditions, concept definitions, classification schemes, cut-points.
- **Coding implementation** will vary based on coder's idiosyncratic features
  - → Perspectives, case familiarity, lived experience, and mood.
- **Downstream use** often treats numeric values as decisive, objective facts, sustaining a "myth of objectivity" (Merry, 2016).

We consider subjectivity as unavoidable. The threat to scientific inquiry arises when dataset creators or downstream users **fail to acknowledge and accommodate it**.

Datasets as collective memory

- Coding entails *thick description* (Geertz, 1973): interpreting political, context-specific, often contested events and then abstracting them into typological categories that permit cross-sectional and -temporal comparison.
- Like archive curation, coding is a process of recording **collective memory**:
  - → **Source material:** Process of representing the past as it was remembered and recorded (Misztal, 2003)
  - → **Annotation:** Collective process of: deciding which events, actors, and perspectives are included (remembered) and how; and deciding which cases are excluded (forgotten).
- The task is not to eliminate human judgment but to surround the data with a **thickly descriptive scaffolding** that records how the memory was constructed.
- **LLM-constructed datasets face similar realities and must also make judgments.**

Datasets and practical interventions

Lab-built conflict datasets

We built three datasets with more than 50 student RAs across three research labs.

Research Lead: Dr. Emily Gade

SAGAA
Syrian Armed Groups, Alliances, and Actors (2012–2018)

More than 28,000 translated militant social media posts, with 21,000 hand-coded attack claims, plus network, hierarchy, and group-level data on 67 organizations.

Troubles on the Ground
Street conflict in Northern Ireland (1971–1973)

Coded from declassified British 72-hour security situation reports at the incident-actor level, covering state forces, paramilitaries, civilians, and informal groups.

Rebel Consolidation
Multi-party insurgencies since 1990

Somalia, Mali, Sri Lanka, Palestine, Liberia, Lebanon, and Afghanistan, coded in three phases: case timelines, group-level attributes, and dyad-level interactions.

Practical interventions

Source of subjectivity	Intervention
Underlying source data are incomplete or contested	Record contestation, missing information, and sources used; flag contested and assumed values; set a minimum search time before coding a zero
Research design (director's positionality)	Reflexivity statements documenting the director's assumptions, priorities, and goals
Coding implementation (coders' positionality)	Coder reflexivity statements; field notes recording judgments and mistakes; discuss and document the nature of coder disagreement
Downstream usage	Publish limitations, source strengths and weaknesses, and the evolution of the ontology; enable sensitivity analyses

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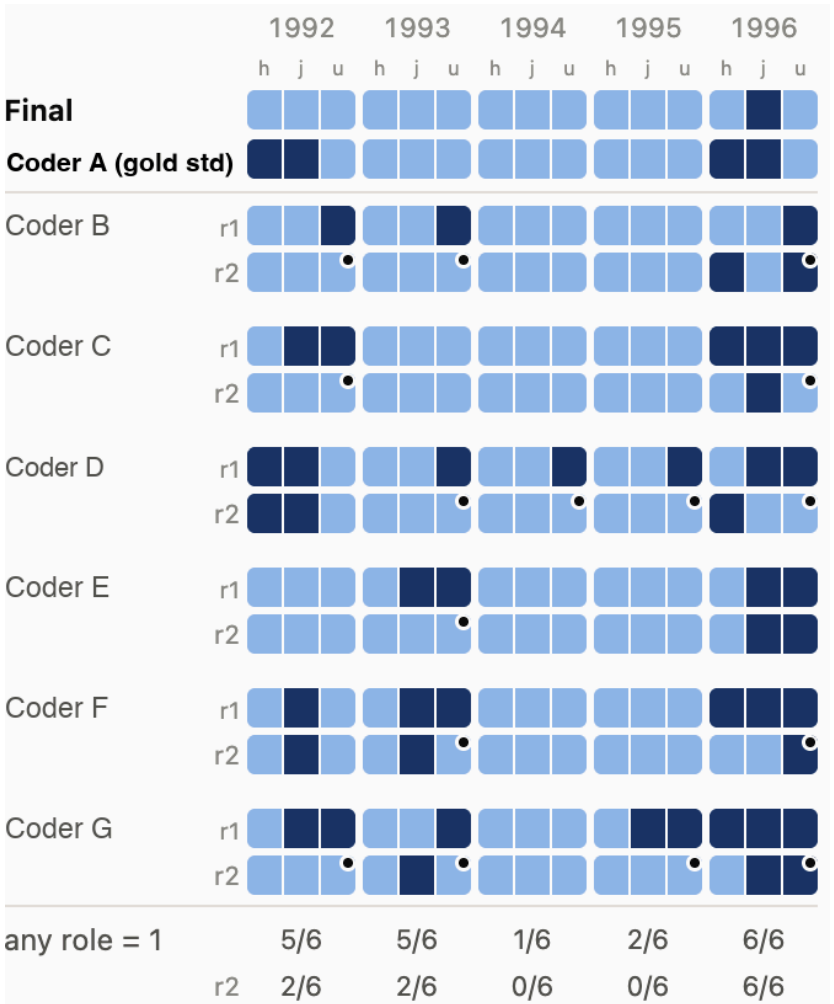
Language models as dataset coders

Can a language model reproduce—or stand in for—the collective human coding process?

Human coding strategy

Pilot coding round: Afghanistan (Rebel Consolidation)

- **Round 1:** Six coders annotated whether Hezb-e Islami, Jamiat-e Islami, or an unknown actor initiated an alliance in each year.
- **Inter-coder discussion:** The team met to clarify the annotation protocol and address questions.
- **Round 2:** Each coder annotated the same years again.
- **Final codes** collectively arrived at following Round 2 discussion.
- *Note*: Annotations still do not achieve 100% agreement.
  - → Discussion reduced misunderstanding but did not remove **differences rooted in interpretation**.
  - → **Record the nature and texture of disagreement** (rather than treating it only as noise).



LLM Audit design

Can a language model reproduce—or stand in for—the collective human coding process?

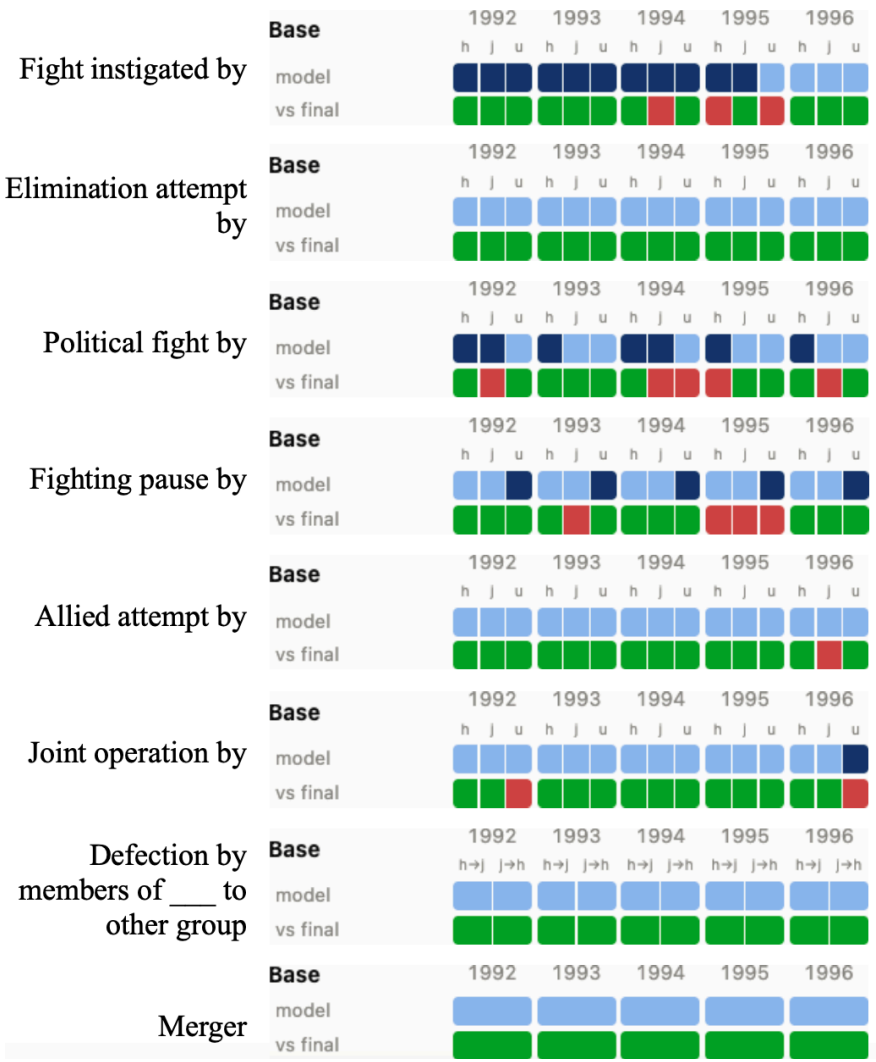
Dyad task: Whether Hezb-e Islami, Jamiat-e Islami, or an unknown actor initiated each of 7 types of interactions (a conflict fight, a political fight, a merger, a defection, etc.) between 1992 and 1996.

Anthropic's Fable 5.1 annotated each variable-year (pooled where the codebook makes decisions interdependent) under three conditions:

Condition	System prompt	Material provided
Base	"Expert on conflict in Afghanistan"	Codebook instructions
Sources-only	"Expert on conflict in Afghanistan"	Codebook + the sources a specific human coder cited
Subjectivity	A 2021 undergraduate research assistant who completed the pre-coding protocol	Codebook + sources + that coder's reflexivity statement (name removed)

Observation #1: Models do not achieve a high level of agreement.

Comparing model responses (base condition) to the final decisions:



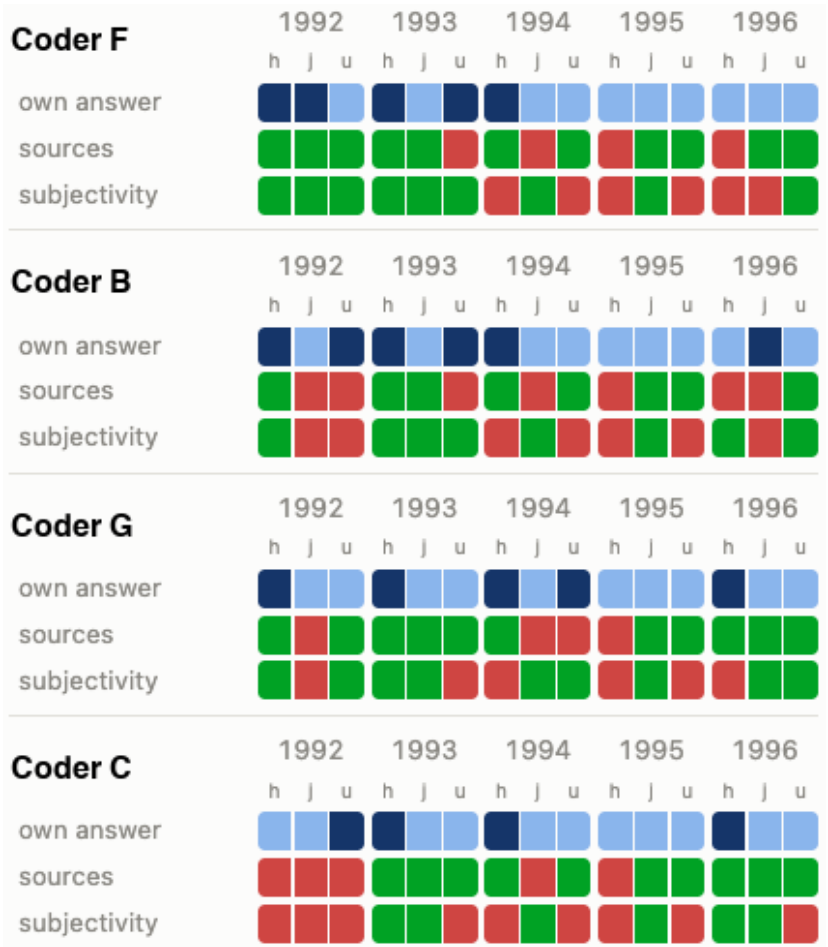
- 15 points of disagreement (dyad cells in red)
- Level of (dis)agreement similar to that achieved by human coders (12-18 points of disagreement)→
- **The model performs similarly to a single human coder.**
- **The model DOES NOT achieve a compelling level of agreement with the collectively finalized codes.**

Observation #2: Supplying sources does not change model decisions.

Condition	System prompt	Material provided
Base	**"Expert on conflict in Afghanistan"**	**Codebook instructions**
Sources-only	**"Expert on conflict in Afghanistan"**	**Codebook + the sources a specific human coder cited**
Subjectivity	A 2021 undergraduate research assistant who completed the pre-coding protocol	Codebook + sources + that coder's reflexivity statement (name removed)

- Across all years, categories, and coders' source sets, the sources-only annotations were **identical** to the base annotations.
- Two possible explanations:
  - → The original model answers **already reflected the human-identified information**.
  - → The model's priors were **entrenched** such that relevant evidence did not move them.
- In theory, if a model is incorporating human-identified material, its decisions may be closer to final human codes.
  - → **Drawing on Observation 1, *maybe* provides evidence for entrenchment (?)**

Observation #3: Reflexivity statements increase model-human disagreement



- In the subjectivity condition, the model received a coder's own reflexive statement about the conflict.
- Reflexive statements changed the model's answers and **increased model disagreement with the coder's round-1 annotations.**
- **Open-ended considerations:**
  - → Presume that conditioning a model to embody a certain perspective could be valuable. *Particularly if seeking to dislodge model from defaulting to mainstream/powerful perspectives.*
  - → What do these findings suggest? Model over-emphasis on positionality? Reasonable human modification or over-correction?
- Baseline effort to inscribe a coder's perspective does not necessarily condition the model to **embody that coder's unique subjective approach to annotation**.

Conclusions

- Constructing large-*n* observational datasets is an inevitably subjective process that should be documented, acknowledged, and considered.
- Specific strategies (e.g., documenting the texture of coder disagreement) make the dataset memory-making process visible and usable to downstream users.
- LLM annotation **cannot reliably emulate the collective human coding process** (Observation 1)
- LLM decisions **may be overly entrenched in baseline training data** (Observation 2)
- Further interrogate the **usefulness of embedding specific subjective perspectives into model behavior** (Observation 3)
- **Next steps?**

Thank you





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References

Geertz, C. (1973). Thick description: Toward an interpretive theory of culture. In *The interpretation of cultures*. Basic Books.

Merry, S. E. (2016). *The seductions of quantification: Measuring human rights, gender violence, and sex trafficking*. University of Chicago Press.

Misztal, B. A. (2003). *Theories of social remembering*. McGraw-Hill Education.